ASSIGNMENT 3

Unit Covered	Unit III — Camera Calibration
Suggested Tools	Python (OpenCV calib3d module), a printed checkerboard, any camera

1. Print a checkerboard pattern (or use a provided one) and capture 12–15 photos of it from different angles and distances using a single camera (phone camera is fine).
2. Run camera calibration to estimate the intrinsic parameter matrix (focal length, principal point) and the radial/tangential distortion coefficients. Report all values clearly in a table.
3. Undistort one of your original images using the estimated coefficients and place the original and undistorted images side by side. Point out, with arrows or annotations, at least two regions where straight real-world lines were visibly curved before correction.
4. For one calibration image, compute and report the extrinsic parameters (rotation and translation of the checkerboard relative to the camera) and explain in your own words what these numbers physically represent.
5. Compare the reprojection error before and after calibration, and explain what an "acceptable" reprojection error would mean for the warehouse's box-measurement use case.
6. Briefly explain, using the orthographic vs. weak-perspective vs. full-perspective camera model concepts, which model best fits a warehouse camera mounted 6 metres above shelves handling boxes of varying height, and why.